# Measuring progress towards the UN Sustainable Development Goal 8: Decent work and economic growth

Charlie Dicker, Haoxuan Du, Achinta Kundra,
Vasiliki Michalia, Jasmine Trabucchi

**The Two Sustainable Development Goals:**

- Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7% gross domestic product growth per annum in the least developed countries.
- By 2020, substantially reduce the proportion of youth not in employment, education or training.

# Introduction

In 2015, the UN created the Sustainable Development Goals (SDGs), which were adopted in 2016 (United Nations). This report will explore how six continents (excluding Antarctica) are faring with respect to two key goals highlighted above, by analysing relevant datasets. It is crucial to understand what these goals aim to achieve. They identify key areas of improvement to support sustainable economic development for both current and future generations. The first goal promotes sustained economic growth across all continents, with a focus on increasing economic well-being in the least developed countries. The second goal strives to better prepare youth to contribute to national development.

In addition to analysing the trends for each SDG between each continent, we analysed trends in certain countries within continents, to further pinpoint contributing factors to net continental trends. It is also important to note that two separate code scripts were written, for the different SDGs.

Overall, we aim to show that progress towards the chosen SDG targets varies significantly across regions, with some continents demonstrating meaningful improvements, and others facing more challenges.

# Methodology

Our analysis included the files provided on the Hub, and additional data including population, youth unemployment and education attainment. Population data enabled us to calculate the weighted average of GDP growth and youth NEET rates for each continent. Youth unemployment and tertiary school enrolment data allowed correlation analysis of these two factors with the NEET. These datasets were standardised and merged to effectively measure the goals.

## Target 1: Growth Targets

### *Sustainable Growth:*

Since the UN created the SDGs in 2015, we only used data between then and 2021.
The following graphs were compiled:

1. Total GDP/Year
   a. A scatter plot and line chart for total GDP against year for each continent observed.
   b. A scatter plot and line chart for total GDP (indexed, starting value 100) against year for each continent observed.
2. GDP Growth (%)/Year
   a. A scatter plot and line chart for mean and weighted (by GDP) economic growth against year.
   b. A scatter plot and line chart for mean economic growth of the continent, and the growth of the 2 countries with the highest GDP against year

### *LDCs 7% Growth:*

The following graphs were compiled:

1. A line chart for economic growth against year for every LDC defined by the UN.
2. A scatter plot and line chart was made for average and weighted economic growth against year for each of the 4 continents with LDCs.
3. A line chart of economic growth against year was made for each LDC which reached over 7% growth during the 2015-2021 period.

## Target 2: youth NEET

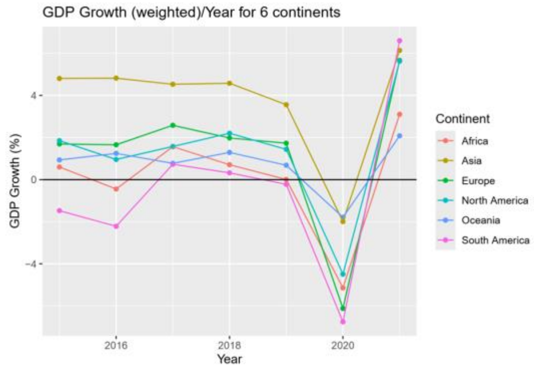
In this section, we mainly used the youth NEET data set. We also created a separate data frame where average values of each indicator for each continent were calculated.
The following graphs and tables were compiled:

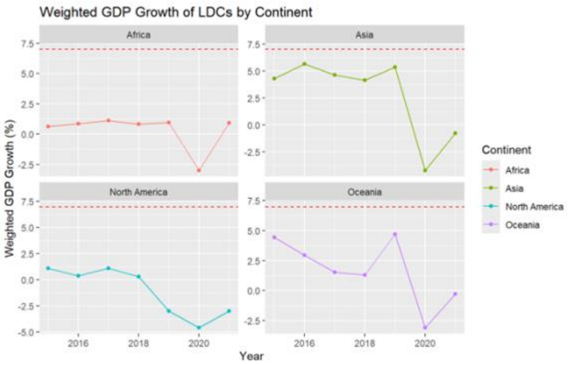
1. A line chart for youth NEET against year to illustrate the change in NEET through the year.
2. A linear regression model for youth NEET rate against year was drawn to illustrate the average change in NEET through the years. A p-test *(5% significance level for all p-test)* was conducted as well to determine if there exist such trend.
3. Two correlation tables, which shows the correlation and p-value of NEET with youth unemployment and school enrolment, continentally and globally respectively. We have used Spearman correlation for these two data sets as they contain outliers and data were not normally distributed.

# Results

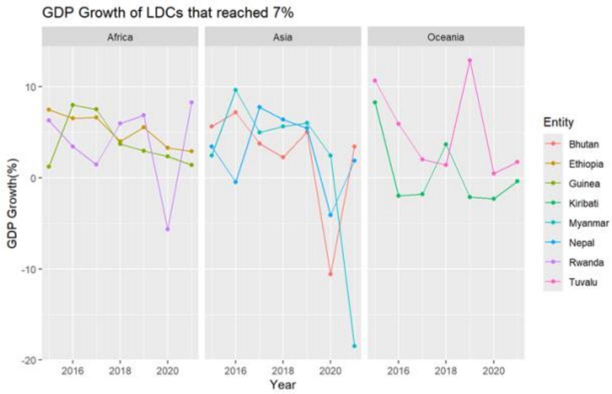
## TARGET 1: per capita economic growth

From the graph showing continents' weighted GDP growth per year, Asia clearly leads, starting at around 5% and maintaining higher growth despite a COVID-19 contraction. Oceania was the least affected by the pandemic, with minor slowdown thanks to low trade reliance and geographic remoteness. Africa shows volatility, with negative growth in 2016, recovery, and a pandemic-induced dip in 2020. Europe had the second-highest growth pre-pandemic, declined in 2020, and recovered by 2021. South America started negative, turned positive in 2017, declined during COVID, and rebounded strongly in 2021.




Turning to LDCs, analysing individual countries is challenging due to high volatility; continental weighted averages provide clearer insight. African LDCs had modest but steady growth, while Asian LDCs suffered sharply during COVID. North American LDCs declined steadily from 2018, and Oceania LDCs also slowed due to reliance on aid.




A 7% growth rate is theoretically attainable for LDCs, but trajectories are unstable. Between 2015–2021, only eight LDCs — Bhutan, Ethiopia, Guinea, Kiribati, Myanmar, Nepal, Rwanda, and Tuvalu — reached this threshold, typically briefly. Examining average growth highlights fast-growing LDCs such as Bangladesh, Ethiopia, Tuvalu, and Rwanda, though even these cases show uneven performance, emphasizing the limitations of annual growth targets as indicators of long-term progress.

## TARGET 2: youth NEET

**Figure 1** illustrates how NEET rates changed across all continents from 1976 to 2020. Asia and Africa displayed some of the highest NEET rates while they also showed significant year-to-year volatility. South America also recorded relatively high NEET rates, but its trajectory was relatively more stable compared to Asia and Africa. On the other hand, Europe, North America and Oceania had the lowest NEET rates and showed relatively stable patterns across time with two exceptions: Oceania experienced a great increase between 2017 and 2020, and North America showed a significant rise between 2019 and 2020.

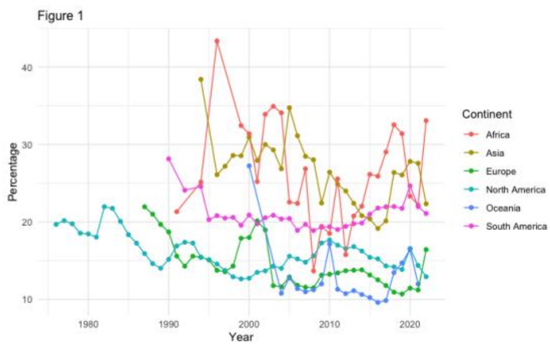
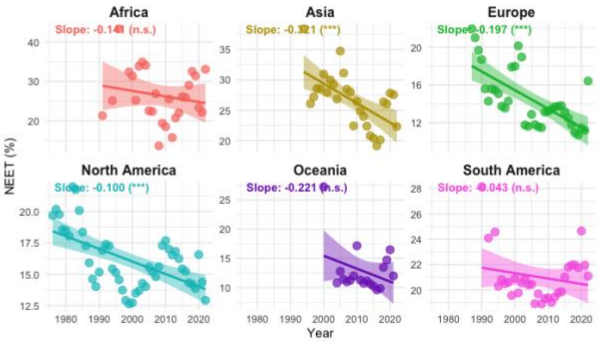



**Figure 2** displays the regression lines for NEET rates over time for each continent. All continents recorded negative regression slopes. However, only three of them showed statistically significant downward trends. Asia demonstrated the greatest decline (-0.321). Additionally, Europe had a moderate decline (-0.197) while North America had a slightly smaller slope (-0.100). In contrast, Africa, Oceania and South America had negative slope coefficients which were not significant. Oceania recorded the second-largest negative slope

Figure 2




(−0.221), although this estimate remained non-significant. Furthermore, Africa showed a smaller negative slope (−0.141) while South America's slope was close to zero (−0.043).

**Table 1** examines the correlations between NEET rates and school enrollment for tertiary education, as well as between NEET rates and youth unemployment for each continent. Starting with NEET and school enrolment for tertiary education, all continents show statistically significant correlations, reflected in extremely small p-values. However, Africa is the only continent with a positive correlation coefficient (0.2232) while it also has the highest p-value (0.0156) among the continents. Furthermore, Oceania has the strongest negative correlation (-0.8927) while Europe (-0.3373) and Asia (-0.3821) have the weakest. Additionally, for NEET and Youth Unemployment, most continents show statistically significant correlations with the exceptions of North America and Oceania. North America shows the weakest correlation (-0.0507) and an extremely high p-value (0.4635). In contrast, Europe (0.6834) and Asia (0.5954) display the strongest corelations and p-values of almost zero.

Table 1: Continental Correlation

| Continent | Corr_SE | P_Value_SE | Corr_Unemp | P_Value_Unemp |
|---|---|---|---|---|
| Africa | 0.2232 | 0.0156 | 0.4211 | 0.0000 |
| Asia | -0.3821 | 0.0000 | 0.5954 | 0.0000 |
| Europe | -0.3373 | 0.0000 | 0.6834 | 0.0000 |
| North America | -0.7692 | 0.0000 | -0.0507 | 0.4635 |
| Oceania | -0.8927 | 0.0000 | 0.2500 | 0.0902 |
| South America | -0.4351 | 0.0000 | 0.5748 | 0.0000 |

Table 2: Global Correlation

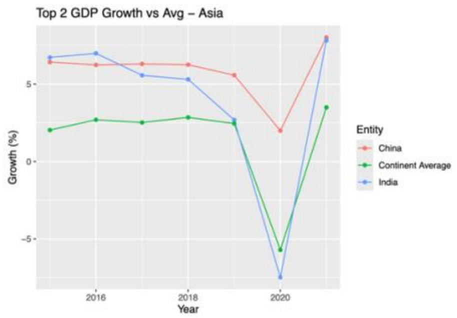
| variable | correlation | p_value |
|---|---|---|
| Unemployment | 0.3161 | 0 |
| SE | -0.5574 | 0 |

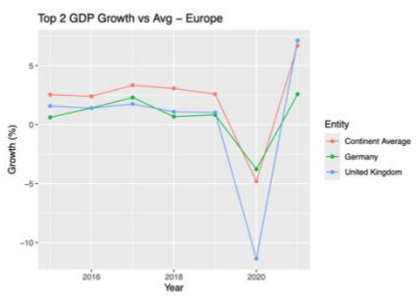
**Table 2** shows the same correlations globally. The results showed a small positive correlation between youth unemployment and NEET rates and a larger negative correlation between school enrolment and NEET rates. Both correlations were statistically significant.

# Discussion

## Target 1

### *Continental trends:*



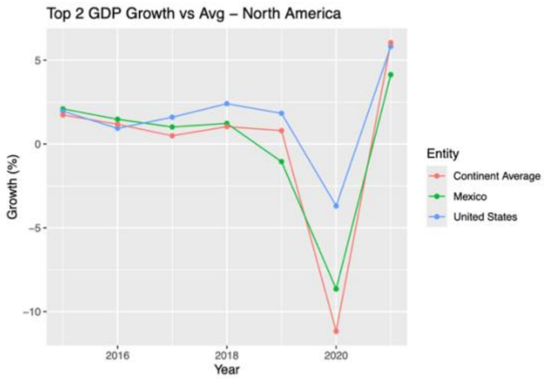

**Asia** had the strongest and most sustained growth pre-covid. China and India were the economic drivers of the continent during this period, as shown by the graph showing Asia, China, and India's growth. China's strong pre-covid growth was most likely driven by heavy infrastructure and urban development spending, as well as being the world manufacturing hub with strong export performance. India has a large, young population, leading to rising consumption and labour supply.

**Africa** had volatile growth, with a small dip in 2016, however it recovered before the pandemic. In several countries, domestic issues such as political instability and conflict may have caused the slight downturn in 2016. Another main driver of this dip was the global fall in commodity prices. Many African economies depend heavily on natural resource exports.

**Europe** had the second highest growth before the pandemic, with a similar pattern to Asia. Unlike Asia however, this growth was not driven by its two largest nations by GDP. The UK, Europe's 2nd largest nation by GDP, suffered much more than Europe on average. This was due to already low growth and productivity levels pre-pandemic, as well as accustoming to the new Brexit trade deals.

**Oceania** had a less severe decline during Covid, most likely due to lower exposure to international economic shocks. New Zealand's approach dealing with Covid was "hard and early". They locked down borders and had early lockdowns. This limited the spread and impact of the virus. Australia rolled out large fiscal support schemes, protecting jobs and stimulating the economy, avoiding a large downturn.

**South America** had low and negative growth leading up to the pandemic, largely due to political instability, inflation, and high levels of poverty. However, South America had the highest growth out of all continents in 2021. One reason may be due to the sharp rebound in global demand; large economies like the US and China had an increased demand for commodities, raw materials and agricultural products post-covid, which are South America's key exports. Exports from Latin America surged: the value of regional goods exports was projected to grow 25% in 2021, after a 20% drop in 2020.




**North America** was hit rather hard by the pandemic, reaching a growth rate of less than -10%. The US however, was not affected as badly, due to fiscal stimulus, vaccination, and strong consumer spending. Mexico recovered more slowly due to more restrictions, and greater trade exposure.

*LDC 7% Growth:*

Although almost reaching 10% in 2016, **Myanmar's** severe economic collapse in 2021 as mentioned in the results was due to the military coup, exacerbating its already fragile economic conditions. Many foreign investors pulled out of Myanmar, due to instability and uncertainty. The military regime also imposed strict controls on foreign currency and restricted imports, leading to cost-push inflation of commodities, increasing the price of many final products.

As mentioned in the results, both **Tuvalu** and **Kiribati** had strong peaks and limited pandemic impact. Their small size and geographic isolation partially insulated them from global economic shocks; however, this same isolation implies a strong dependence on international aid. For instance, the Tuvalu Trust Fund, established in 1987 by the United Kingdom, Australia and New Zealand, has played a key role in supporting the country's economy, though such financial assistance is often volatile and unpredictable. Both countries rely heavily on revenues from fishing licences, yet environmental challenges constrain the full exploitation of this resource. Moreover, their remoteness delayed the spread of COVID-19: Tuvalu only recorded its first cases in November 2022, while Kiribati began registering cases in early 2022.

*Note: Tuvalu graduated from being an LDC in late 2021, hence the strong figures*

## Target 2

**Africa** displays an insignificant negative slope in NEET over time, indicating inconsistent progress towards the target. Despite the implementation of policies such as Jobs for Youth in Africa, political instability in North Africa and changes in social dynamics led to the fluctuating NEET rate (Ibourk and Elouaourti, 2025), which also explained the weak correlation of NEET with both school enrolment and youth unemployment. Furthermore, the great informal sector may lead to overestimation of NEET level of the continent (Cieslik et al., 2021).

A similar trend was also observed in **South America**. Despite the policy progress toward eliminating NEET, including the Youth in Action programme in Colombia and the National Youth Inclusion Programme in Brazil, factors including a large informal sector, adolescent pregnancy and early education dropout, especially for young women, all contribute toward the substantial NEET (CD/CAF/ECLAC, 2016).

**Asia** shows a significant negative trend, implying a generally consistent progression towards the target despite the rise after 2016. This recent rise in NEET is likely driven by rising structural unemployment, possibly linked to the rapid mechanisation of key sectors. (Schipper et al., 2025), which explains the correlation between NEET and unemployment. Additionally, the missing NEET rates for China and Korea suggest that the trend was disproportionately shaped by countries in South and Southeast Asia, thereby introducing bias in our results.

**Europe** shows a stable trajectory especially in recent years and a significant negative trend, indicating that the continent continuously improves in terms of the target. The stronger correlation between youth unemployment and NEET compared with school enrolment aligns with the impact of labour market policies such as the Youth guarantee which has significantly reduced the NEET rates since its establishment in 2013, by minimizing the discouraged-worker effect. (European Commission, 2024)

**North America** shows a statistically significant downward trend in NEET despite fluctuations. The data for North America is largely dominated by US data since the US takes the most population. Several government policies were implemented, including the Workforce Innovation and Opportunity Act of 2014, which primarily focuses on disadvantaged youth through vocational education and training (U.S. Department of Labor, n.d.), which reduced the NEET in the last decade by 5%.

Similar policies were implemented by **Oceania** countries like Australia, where the Work for Dole policy provided work-like experience and incentive to work (Department of Education, Skills and Employment, 2022), and New Zealand, where the Youth Guarantee Fund provides fee-free tertiary education to disadvantaged youth (Tertiary Education Commission, 2024). These policies effectively kept NEET at a low stable level and explained the strong correlation between NEET rate and tertiary school enrolment rate.

# Conclusion

Overall, despite global progress towards the two SDGs, the pace and distribution of these improvements remain uneven, and concentrated in certain regions. As mentioned, the imbalance can be understood when examining certain infrastructural changes occurring rapidly in some regions, and at slower rates in others.
First focusing on target one, Asia demonstrates key success, as the continent has the most sustained pre-pandemic GDP growth rate, and the highest rate during the pandemic, largely driven by China and India, whereas Africa and South America experienced more volatility due to external shocks.
For least developed countries, achieving the 7% annual growth target proved possible but not consistent, as indicated by the large fluctuations in growth rates in these countries. Upon analysis, this was likely due to political and economic stability.
Moving onto target two, reducing youth NEET rates also showed divided outcomes. Europe, North America and Oceania achieved more meaningful progress towards the target, due to labour market and education policies implementations.
Interestingly, all continents apart from Africa achieved a net decrease in NEET rates, highlighting global progress. However, this progress again comes inconsistently, reflecting persistent obstacles such as political instability and barriers to education access. Correlation analysis highlighted that reductions in NEET rates are closely associated with higher tertiary education enrolment and lower youth unemployment, demonstrating the importance of coordinated policy aimed at both education and labour market access.

Progress towards these two SDGs is possible but highly sensitive to national circumstances and global shocks, as seen throughout history. Looking ahead, achieving the goals will require global investment into employment pathways, education and training, and systems to protect against future global crises.

Additionally, one may note that progress towards these goals also have their own associated issues. For example, sustained levels of GDP growth may cause a trade-off with sustainability, another important consideration in the United Nation's SDGs.

## Reference


BBC News. (2012) 'Kiribati country profile', BBC News, 22 March, https://www.bbc.co.uk/news/world-asia-pacific-16431122

Carcillo, S. et al. (2015), "NEET Youth in the Aftermath of the Crisis: Challenges and Policies", *OECD Social, Employment and Migration Working Papers*, No. 164, OECD Publishing, Paris, https://doi.org/10.1787/5js6363503f6-en

Cieslik, K., Barford, A. and Vira, B. (2021) 'Young people not in Employment, Education or Training (NEET) in Sub-Saharan Africa: Sustainable Development Target 8.6 missed and reset', *Journal of Youth Studies*, 25(8), pp. 1126–1147. doi:10.1080/13676261.2021.1939287

Collins, B. and Adam, E. (2022). *Youth Employment and Training Programs Authorized by the Workforce Innovation and Opportunity Act*. [online] Congress.gov, https://www.congress.gov/crs-product/IF11952

Department of Education, Skills and Employment (2022). *Work for the Dole General Fact Sheet*. [online], https://www.dewr.gov.au/work-dole/resources/work-dole-general-fact-sheet

ECLAC (2021) International Trade Outlook for Latin America and the Caribbean, https://www.cepal.org/sites/default/files/presentation/files/211207_eng_final_ppt_internaciona_trade_outlook_2021.pdf

European Commission (2024). *The reinforced Youth Guarantee*. [online] Employment, Social Affairs and Inclusion, https://employment-social-affairs.ec.europa.eu/policies-and-activities/eu-employment-policies/youth-employment-support/reinforced-youth-guarantee_en

Ibourk, A. and Elouaourti, Z. (2025) *Youth 'Not in Education, Employment, or Training' (NEET) in North Africa: Do Household Structure, Subjective Factors, and Political Dimensions Affect NEET Outcomes Differently by Gender?* [pdf] ERF 31st Annual Conference, https://erf.org.eg/app/uploads/2025/03/1742721067_153_712761_erf31st_pe121_aomaribourk_zakariaelouaourti.pdf

OEC. (n.d.) *Kiribati: Country Profile, https://oec.world/en/profile/country/kir*

OECD/CAF/ECLAC (2016), *Latin American Economic Outlook 2017: Youth, Skills and Entrepreneurship*, OECD Publishing, Paris, https://doi.org/10.1787/leo-2017-en

Schipper, S., Crowley, S. and Zuniga, B.E.T. (2025). *How to Solve Asia's Youth Employment Crisis | Asian Development Blog*. [online] Adb.org, https://blogs.adb.org/blog/how-solve-asia-s-youth-employment-crisis

Tertiary Education Commision. (2024). *Youth Guarantee Fund*. [online], https://www.tec.govt.nz/funding/fund-finder/youth-guarantee

U.S. Department of Labor (n.d.). *Youth Programs*. [online], https://www.dol.gov/general/topic/training/youth

United Nations, The Sustainable Development Agenda, https://www.un.org/sustainabledevelopment/development-agenda-retired/

Wikipedia (n.d.) *Economy of Tuvalu,* https://en.wikipedia.org/wiki/Economy_of_Tuvalu

World Bank (2021) *Boost of Tuvalu's economic, social and climatic resilience,* https://www.worldbank.org/en/news/press-release/2021/12/08/boost-for-tuvalu-s-economic-social-and-climate-resilience

World Bank (2025a). *Population, Total*. [online] World Bank Open Data, https://data.worldbank.org/indicator/SP.POP.TOTL

World Bank (2025b). *School enrollment, secondary (% gross)* . [online] World Bank Open Data, https://data.worldbank.org/indicator/SE.SEC.ENRR

World Bank (2025c). *Unemployment, Youth Total (% of Total Labor Force Ages 15-24) (modeled ILO estimate) | Data*. [online] World Bank Open Data, https://data.worldbank.org/indicator/SL.UEM.1524.ZS